YafaRE at KnowledgeGraphEval 2026: A Relation Extraction ArBERTv2-based Approach for Arabic Knowledge Graph Construction

Anas Melhem[1], Osama Hamed[1,2,*], Noora Shawareb[1]

[1]Palestine Technical University - Kadoorie, Palestine
[2] Forschungszentrum Jülich, Germany
{a.melhem, osama.hamed, noora.shawareb}@ptuk.edu.ps
***Correspondence:** osama.hamed@ptuk.edu.ps

Abstract

Relation extraction (RE) identifies semantic relations between entities, converting unstructured text into structured facts for downstream applications such as knowledge graph construction. Arabic RE remains challenging due to rich morphology, complex syntax, dialectal variation, and the scarcity of high-quality annotated resources. We present Yafa's Team submission (ranked 7th) to Subtask 2 of the KnowledgeGraphEval 2026 Shared Task, which frames Arabic RE as multiclass classification over 40 relation types from the *Wojood*^{Relations} dataset. We propose an ArBERTv2-based entity-marker classifier that fuses global sentence context ($[CLS]$) with entity-specific representations (h_e1, h_e2) into a joint representation for relation prediction. Building on this baseline, we exploit a self-training step that pseudo-labels the released test inputs at a softmax confidence threshold of 0.95 and retrains on them, obtaining Micro-F1 of 91.92%. We further conduct a systematic analysis of cross-backbone ensembling with MARBERTv2 and AraBERTv2, finding that ensembling does not outperform the single model. Our results highlight self-training as a simple yet effective strategy for improving Arabic RE robustness.

1 Introduction

Relation extraction (RE) identifies semantic relations between entities in text. It is a fundamental information extraction task because it turns unstructured sentences into structured facts that can be reused by downstream NLP systems. These facts are especially useful for constructing and enriching knowledge graphs, where entities and relations are organized into machine-readable representations (Aljabari et al., 2025b,a). Arabic NLP remains challenging because Arabic exhibits rich morphology, complex syntax and semantics, and wide variation across writing styles and dialects (Hamed, 2019). Consequently, Arabic RE also suffers from a scarcity of high-quality annotated resources.(Aljabari et al., 2025b). Existing Arabic resources often cover only narrow relation sets or appear as small Arabic portions within multilingual corpora, which restricts robust model development and evaluation (Aljabari et al., 2025b,a). These challenges motivate the need for robust Arabic RE systems that can better support knowledge graph construction in realistic settings.

This paper presents Yafa's Team submission (ranked 7th on the leaderboard) to Subtask 2 of the KnowledgeGraphEval 2026 Shared Task (Aljabari et al., 2026) for Arabic RE, focusing on extracting semantic relations between entities in Arabic text. The shared task frames relation extraction as a multiclass classification problem: given a sentence and a pair of entities, the objective is to identify the semantic relation that holds between them. This task is based on *Wojood*^{Relations}, a large-scale Arabic dataset annotated with 40 predefined relation types covering diverse semantic categories (Aljabari et al., 2025b). To address this task, we develop an ArBERTv2-based entity-marker classifier for Arabic relation extraction. Subject and object entities are explicitly marked within each sentence using [E1]...[/E1] and [E2]...[/E2] tokens, and the model builds a joint representation by concatenating the global sentence representation ([CLS]) with the entity-specific hidden states (h_e1, h_e2), which is then passed to a classification layer over the 40 relation types. This raises the micro-F1 score from 0.702 to 0.9192. Overall, this paper makes three contributions:

1. We present an ArBERTv2-based relation extraction model with an entity-marker architecture that classifies each entity pair from a concatenation of the [CLS] token representation (global sentence context) and the hidden representations of the two marked entities (h_e1,

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h_e2).

2. We introduce a semi-supervised transductive self-training step that pseudo-labels the released test inputs (softmax ≥ 0.95) and retrains on them.

3. We provide a systematic analysis of cross-backbone ensembling (MARBERTv2 and AraBERTv2), showing that it does not outperform the single transductive model.

The paper is organized as follows: Section 2 presents prior and recent research on Arabic RE. Section 3 provides an overview of the datasets used. Sections 4 and 5 describe the proposed systems and experimental setup. Section 6 presents and discusses our experimental results. Finally, Section 7 concludes the paper and suggests directions for future work.

2 Background

In recent years, language models have driven substantial progress in RE, enabling systems to move beyond surface-level lexical patterns and capture deeper contextual and semantic relationships between entities. At the sentence level, one common approach marks the subject and object spans with entity markers or special tokens, then classifies the relation from the encoder's contextualized representations. DialogRE (Yu et al., 2020) applies this to dialogue text, marking subject and object boundaries with special tokens. Yan et al. (2022) compare the pipeline and joint approaches to Entity and Relation Extraction and find that the best joint models beat the best pipelines.

SRED-FM and RED-FM (Huguet Cabot et al., 2023) are companion multilingual relation extraction datasets built from Wikipedia using the cRocoDiLe tool and mBART to normalize relation types across 18 languages. SRED-FM is the larger, automatically generated set, with 40 million relations total; its Arabic subset holds 2.9 million relations across 393 of the dataset's 400 relation types. RED-FM is smaller and manually revised, used for evaluation. Both feed mREBEL (Huguet Cabot et al., 2023), a sequence-to-sequence RE model trained on SRED-FM that became the state-of-the-art baseline for later Arabic RE work.

Rakan Al Mraikhat et al. (2024) build on this with AREEj, which augments the Arabic subset of SRED-FM with evidence annotations. Text spans justifying a predicted relation are included. A relation and its evidence are also predicted together. Since SRED-FM has no evidence labels, the authors generate them with ChatGPT-4 using chain-of-thought prompting (Wei et al., 2022), fine-tune an mBART-based model (ArBartEv) on those, then use ArBartEv to annotate all 2.9 million Arabic relations. AREEj beats mREBEL despite far fewer relations (2.9 million versus 40 million): 0.72 F1 on extraction against 0.69, and 0.78 versus 0.71 on classification. It catches 72 relations mREBEL misses; mREBEL catches 47 it misses. Evidence raises recall substantially at a small cost to precision, which the authors attribute to evidence grounding predictions in the text.

This mirrors a broader pattern: using an LLM to bootstrap silver labels for a smaller task-specific model, rather than fine-tuning the LLM directly. Wadhwa et al. (2023) find chain-of-thought prompting consistently helps GPT-3 and Flan-T5 on RE.

3 Dataset

This section presents an overview of the *Wojood*^{Relations} dataset (Aljabari et al., 2025b), released by the shared task organizers for Subtask 2, and reports descriptive statistics on its structure and label distribution.

3.1 *Wojood*^{Relations} Corpus

Wojood^{Relations} is a large-scale Arabic dataset for semantic relation extraction that extends the original Wojood named entity recognition (NER) corpus (Jarrar et al., 2022) with annotated semantic relations between entities. Each instance is represented as a ternary tuple (subject, relation, object), drawn from 40 relation types spanning diverse semantic categories, such as `Location.located_in`, `PartOf.geopolitical_division`, and `Personal.has_occupation`, along with a `no_relation` label for entity pairs that co-occur in a sentence without holding any of the predefined relations.

This dataset consists of 19,455 observations, 17,381 of which are used for training and 2,074 for validation. The training set is dominated by the `no_relation` label and exhibits a highly unbalanced distribution across the 40 relation categories, with the top 15 categories (including `no_relation`) accounting for approximately 98% of the training data. To keep our internal validation stable and reduce noise from extremely rare classes,

we restricted our development-set analysis to these 15 categories; the official evaluation, in contrast, is performed over all 40 relation types. Figure 1 illustrates the relation distribution.

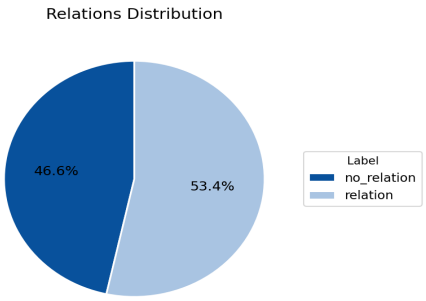


Figure 1: Distribution of relations in the training set.

4 System Description

This section describes the system developed for Subtask 2 of the Arabic RE Shared Task. The code and released resources are available at https://github.com/ohamed/Arabic-RE.

4.1 Model Architecture

We adopt an entity-marker architecture for relation classification. Given a sentence containing a subject and an object entity, the two entity spans are first delimited in the raw text using special marker tokens, [E1]...[/E1] for the subject and [E2]...[/E2] for the object, before the sentence is passed through the ArBERTv2 encoder. This markup allows the encoder to condition its contextual representations on the identity and position of each entity within the sentence, rather than relying on separately encoding entity mentions outside their sentential context.

From the encoder's output, we extract three vectors: the hidden state at the [CLS] token, which summarizes the sentence as a whole, and the hidden states at the [E1] and [E2] marker positions, denoted h_e1 and h_e2, which capture entity-specific contextual information for the subject and object, respectively. These three 768-dimensional vectors are concatenated into a single 2304-dimensional pair representation, $\mathrm{concat}([\mathrm{CLS}], h_{e1}, h_{e2})$, that jointly encodes global sentence semantics and the two entities to be related.

This representation is passed through a classification head consisting of a linear projection down to 768 dimensions, a $tanh$ non-linearity, a dropout layer for regularization, and a final linear layer projecting to 40 output units, one per relation type (including no_relation): $\mathrm{Linear}(2304 \rightarrow 768) \rightarrow \mathrm{Tanh} \rightarrow \mathrm{Dropout} \rightarrow \mathrm{Linear}(768 \rightarrow 40)$. A softmax over these 40 logits yields the predicted relation distribution for the entity pair.

We further extend our method using semi-supervised transductive self-training. First, we use the best inductive model, which was trained on the official training set, to generate pseudo-labels for the released unlabeled test instances. We retain only the high-confidence predictions (softmax $\geq$ 0.95), resulting in 3,758 pseudo-labeled instances out of 4,386 candidate pairs. Then, the model is retrained on the original training set, which is augmented with these pseudo-labeled instances. This approach achieves 91.92% Micro-F1 official test. Figure 2 in Appendix A illustrates the full architecture.

4.2 Cross-backbone Ensemble

We further investigate cross-backbone ensemble methods that combine ArBERTv2 with MARBERTv2 (Abdul-Mageed et al., 2021) and AraBERTv2 (Antoun et al.). Specifically, we test: (i) two-model probability averaging (ArBERTv2 + MARBERTv2), reaching 91.64%; (ii) three-model averaging (ArBERTv2 + MARBERTv2 + AraBERTv2), reaching 91.69%; (iii) a confidence-gated selective merge that overrides ArBERTv2 only where it is unconfident and the two auxiliary models strongly agree, which does not exceed the base model; and (iv) soft low-weight averaging (weights 0.85/0.08/0.07), reaching 91.72%. None of these variants exceeds the single transductive ArBERTv2 model (91.92%), since the auxiliary backbones are individually weaker on this data and ArBERTv2 and MARBERTv2 already agree on 94.3% of test pairs; on the remaining disagreements, neither auxiliary model is reliably more correct, so averaging lowers rather than raises the overall score.

5 Experimental Setup

All experiments are conducted on the Google Colab platform, for both training and inference. Table 2 in Appendix A outlines the main hyperparameters used for fine-tuning the ArBERTv2 model.

6 Results

This section presents our experimental results, ranked by Micro-F1. Table 1 compares our method against the BERT baseline (ArBERTv2).

Our method obtains 91.92% Micro-F1, compared to 88.61% for the baseline. This improvement is driven primarily by precision: our model reaches 96.19%, substantially higher than the baseline, while recall (88.01%) remains close to, and marginally below, the baseline's 88.65%. This indicates that self-training makes the model more accurate in the relations it predicts, reducing false positives, rather than increasing the number of true relations it successfully identifies.

Model	Prec. (%)	Rec. (%)	Micro F1 (%)
Baseline (ArBERTv2)	–	–	93.09
LLM baseline (gpt-4o)	89.25	83.66	85.78
ArBERTv2-T	**96.19**	88.01	91.92

Table 1: A Performance comparison of used models.

6.1 Error Analysis

The results in Table 1 mean that the false-positive rate among predicted relations is about 3.81%. Also, the number of false negatives is roughly 3.4 times the number of false positives. To better understand the remaining errors after transductive self-training, we analyze the predictions of our final ArBERTv2 model on the development set, grouping the observed errors into three categories.

Confusion with `no_relation`. Given the severe class imbalance in the training data (Section 3), a substantial share of errors involve the `no_relation` label. These errors occur in two directions: cases where the gold label is `no_relation` but the model predicts a specific relation type (false positives), and cases where a true relation is missed and predicted as `no_relation` (false negatives). This pattern is consistent with our precision/recall results (Section 6). The model achieves high precision, indicating that its specific relation predictions are generally reliable, while the lower recall indicates that some true relations remain unidentified.

Confusion between semantically adjacent relations. The remaining errors include confusions between relations that are semantically close. Such confusions arise when different relation types share similar lexical or contextual cues. For example, relations describing different types of affiliation, location, or organizational relationships may require distinctions that are not explicit in the local context. These cases suggest that the model can have difficulty distinguishing fine-grained semantic relations when their surface contexts are similar.

7 Conclusion

In conclusion, this paper presents the ARBERTv2 transductive (ARBERTv2-T), a robust Arabic relation extraction approach based on the ArBERTv2 model, which improves the performance of Arabic RE systems. Our team, dubbed Yafa's Team (YafaRE is ranked 7th), achieved a micro-F1 score of 91.92% on the WojoodRelations benchmark. This score outperforms the LLM baseline and is comparable to the ArBERTv2 baseline model. The results demonstrate the effectiveness of our semi-supervised transductive self-training approach, which leverages pseudo-labeled data to improve the model's generalization capabilities. The experiment also explores the use of cross-backbone ensemble methods. However, these methods do not outperform the single transductive ArBERTv2 model due to its strong individual performance and high prediction agreement. The study provides a reliable and fine-grained Arabic RE model that generalizes well across domains and relation types, addressing the limitations of existing models. Future work will examine the feasibility of using other Arabic pretrained language models or small LLMs for this task.

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A Additional Experimental Details

This appendix is meant to provide further details that are essential for understanding our system description paper. This include, among others, the ArBERTv2 entity-marker architecture, (Figure 2), the hyperparameters used for fine-tuning ArBERTv2 (Table 2), and the limitation of this sStudy .

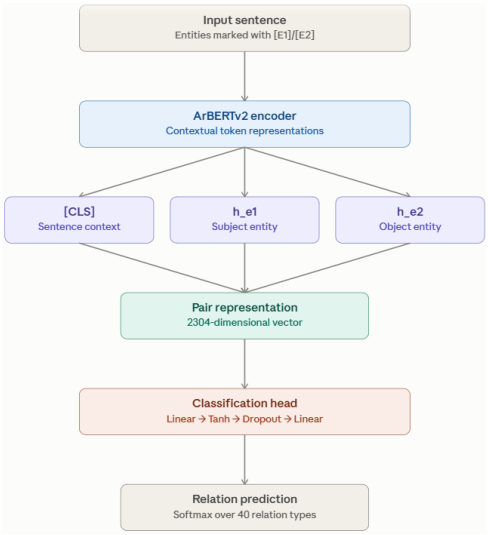


Figure 2: The ArBERTv2 entity-marker architecture used for the relation classification subtask.

Hyperparameter	**Value**
Learning rate (LR	2×10^{-5}
Weight decay	0.01
Batch size (train/eval)	16
Maximum sequence length	160
Number of epochs	4
Linear scheduler warmup	10%
Optimizer	AdamW
Pseudo-label confidence	0.95

Table 2: Main hyperparameters used for fine-tuning ArBERTv2.

Study Limitations One limitation of this work is that the WojoodRelations test set was unlabeled during system development. Therefore, direct error analysis on the target-domain predictions was not possible. Our final system relies on transductive self-training using the released test inputs. This improves robustness under distribution shift, but it also ties the model to the specific test set, rather than yielding a fully inductive model. Additionally, the experiments were conducted using ArBERTv2-based models. While we explored MARBERTv2 and AraBERTv2 in our ensembling analysis, we did not evaluate other Arabic pretrained language models.